PROJECT COLLECTION

REZA BABAPIRI

Automotive & Vehicle Simulation

Vehicle models, dynamic systems and co-simulation

5 TECHNICAL CASE STUDIES

Engineering scope / Method / Results / Visual evidence / Limitations

Technical design, coding and analysis: Reza Babapiri

reza-babapiri.github.io

ADVANCED DRIVER ASSISTANCE · CONTROL

Adaptive Cruise Control | PID vs Observer

Distance-error regulation and disturbance estimation

ENGINEERING QUESTION

How should following-distance feedback be organized when lead-vehicle acceleration and disturbances are not directly measured?

ROLE & CONTRIBUTION

I developed and compared PID-based and observer-based adaptive cruise control approaches using analytical models and MATLAB simulation.

ENGINEERING DECISIONS

- Formulated the longitudinal distance-error dynamics with leader behaviour and disturbances.
- Designed a conventional PID controller as an accessible reference strategy.
- Evaluated a state-observer-based control formulation and compared tracking and robustness behavior.

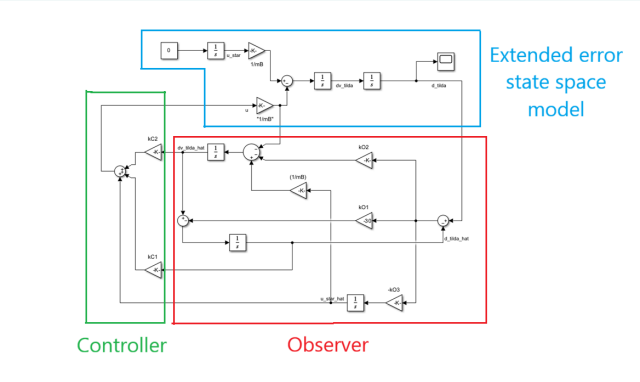
SELECTED ENGINEERING EVIDENCE

Coursework provides time-domain response figures for the controller alternatives and error trajectories; outputs are simulation-only.

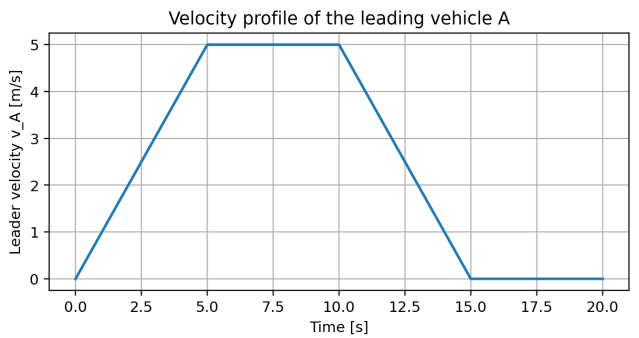
VISUAL EVIDENCE · ADVANCED DRIVER ASSISTANCE · CONTROL

Adaptive Cruise Control | PID vs Observer

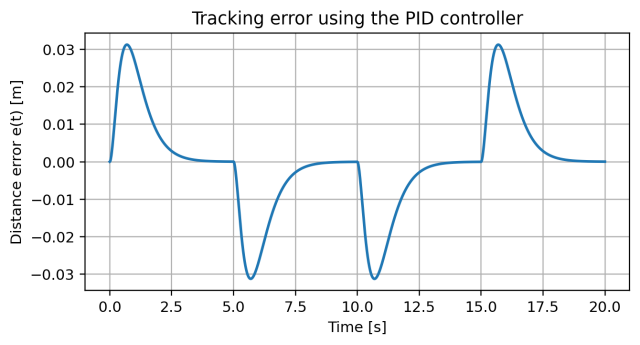
Original images, models and results from the supplied project material



Adaptive cruise-control dynamic response and design comparison.



Comparison plot from observer-based and PID control study.



Distance-error tracking in the closed-loop controller simulation.

RESULTS & ENGINEERING INTERPRETATION

Coursework provides time-domain response figures for the controller alternatives and error trajectories; outputs are simulation-only.

VEHICLE DYNAMICS · STATE SPACE

BMW 730 | Steering Dynamics

Speed-dependent stability, damping and steering response

ENGINEERING QUESTION

How do vehicle forward speed and model parameters change lateral stability and steering-response behavior?

ROLE & CONTRIBUTION

I analysed the specified BMW 730 bicycle-model problem, assessed dynamic properties and implemented feedback studies in MATLAB.

ENGINEERING DECISIONS

- Computed speed-dependent poles/eigenvalues and transfer-function responses for the academic lateral vehicle model.
- Studied controllability, observability and understeer/oversteer tendencies.
- Analysed the influence of feedback design on damping and transient response.

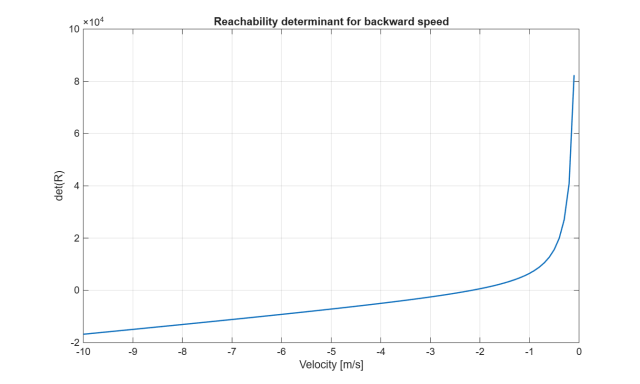
SELECTED ENGINEERING EVIDENCE

The report shows a selected damping-ratio change from roughly 0.088 to 0.628 under its model and controller assumptions.

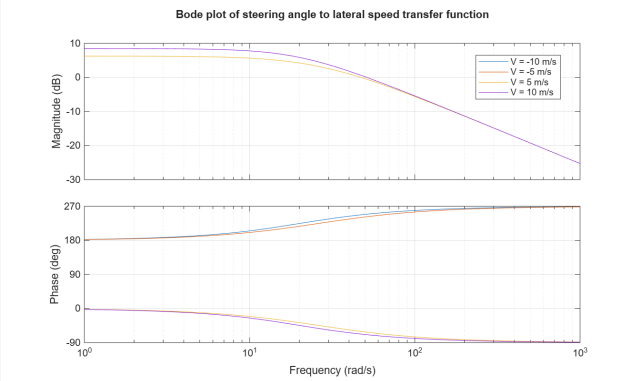
VISUAL EVIDENCE · VEHICLE DYNAMICS · STATE SPACE

BMW 730 | Steering Dynamics

Original images, models and results from the supplied project material



BMW steering-model response analysis derived from the academic model.



Frequency-domain analysis of lateral / steering dynamics.

RESULTS & ENGINEERING INTERPRETATION

The report shows a selected damping-ratio change from roughly 0.088 to 0.628 under its model and controller assumptions.

STATE ESTIMATION · VEHICLE CONTROL

Active Steering | Observer & Wiener Filter

PI feedback versus observer-based yaw-rate regulation

ENGINEERING QUESTION

Can active steering regulation and state estimation remain useful when lateral vehicle dynamics are uncertain or noisy?

ROLE & CONTRIBUTION

I developed steering-controller and observer comparisons, including PI feedback, Luenberger observation and covariance-weighted estimation.

ENGINEERING DECISIONS

- Built a closed-loop yaw-rate/steering formulation from the selected vehicle model.
- Derived PI-controller parameters with characteristic-equation objectives.
- Compared deterministic observer behavior with Wiener/Kalman-like covariance-weighted estimation assumptions.

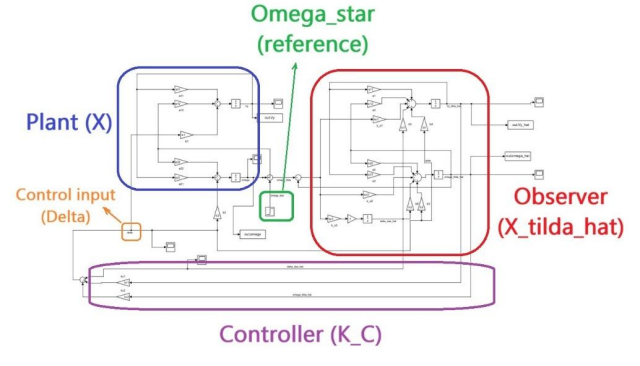
SELECTED ENGINEERING EVIDENCE

The supplied report includes observer/controller diagrams and sensitivity plots; no numerical road-test claim is made.

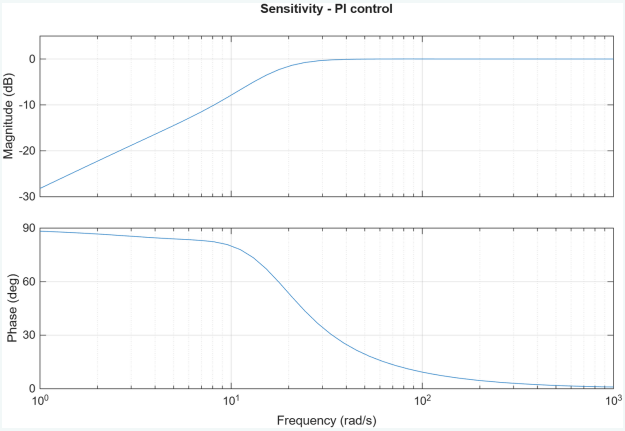
VISUAL EVIDENCE · STATE ESTIMATION · VEHICLE CONTROL

Active Steering | Observer & Wiener Filter

Original images, models and results from the supplied project material



Active-steering controller or observer architecture from the analysis.



Sensitivity and closed-loop behavior in the course simulations.

RESULTS & ENGINEERING INTERPRETATION

The supplied report includes observer/controller diagrams and sensitivity plots; no numerical road-test claim is made.

DYNAMIC SYSTEMS · BALANCE CONTROL

Bicycle Roll | Stability & Feedback

Speed-dependent roll dynamics and feedback stabilization

ENGINEERING QUESTION

Why does balancing performance change with forward speed, and what type of feedback can regulate bicycle roll?

ROLE & CONTRIBUTION

I modelled bicycle roll/steering dynamics, tested stability assumptions and studied candidate feedback controllers in MATLAB.

ENGINEERING DECISIONS

- Used simplified roll equations and a Whipple-type bicycle model to study a balance-control problem.
- Inspected poles, zeros and steering-to-roll dynamics at different speeds.
- Compared the behaviour of elementary proportional control with more structured feedback.

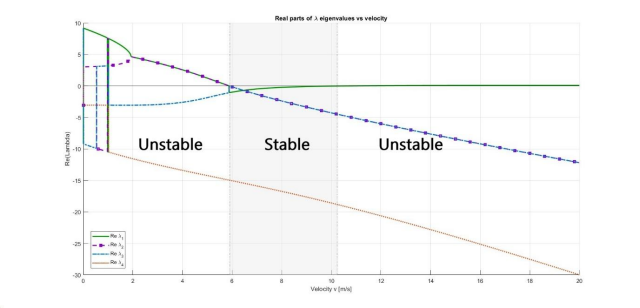
SELECTED ENGINEERING EVIDENCE

The report documents transfer functions, stability plots and controller comparisons under the adopted academic model assumptions.

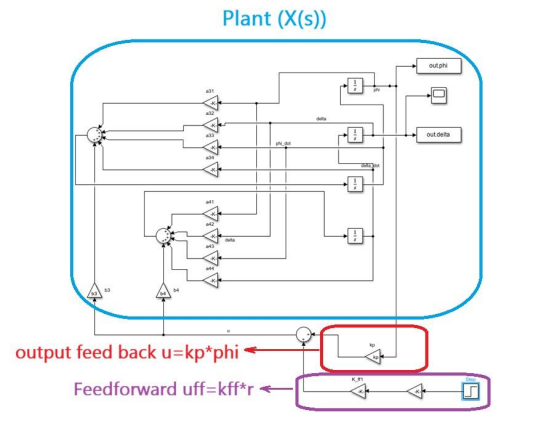
VISUAL EVIDENCE · DYNAMIC SYSTEMS · BALANCE CONTROL

Bicycle Roll | Stability & Feedback

Original images, models and results from the supplied project material



Bicycle roll or steering stability plot from the model analysis.



Feedback system for bicycle-dynamics stabilization.

RESULTS & ENGINEERING INTERPRETATION

The report documents transfer functions, stability plots and controller comparisons under the adopted academic model assumptions.

AUTOMOTIVE SIMULATION · PYTHON

EV / HEV Traffic Co-Simulation | Rome

SUMO traffic + Python bridge + MATLAB/Simulink vehicle models

ENGINEERING QUESTION

How can urban traffic demand be connected to electric and hybrid vehicle model simulations to compare speed tracking and energy use?

ROLE & CONTRIBUTION

I prepared the SUMO urban-route simulation, Python-to-MATLAB/Simulink interface and archived EV/HEV scenario outputs.

ENGINEERING DECISIONS

- Defined different urban traffic intensities and exported driving profiles from SUMO scenarios.
- Linked speed profiles to EV and HEV dynamic models through Python and MATLAB/Simulink scripts.
- Compared reference versus simulated speed, battery charge trends and model energy outputs over traffic cases.

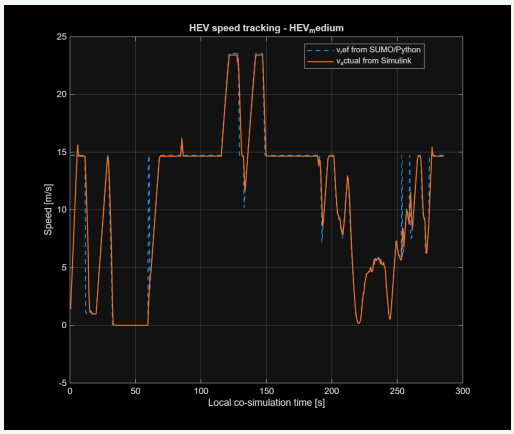
SELECTED ENGINEERING EVIDENCE

The supplied archive contains SUMO inputs, Python scripts, Simulink model files, scenario CSV tables and plotted results. Archived outputs were inspected but not rerun.

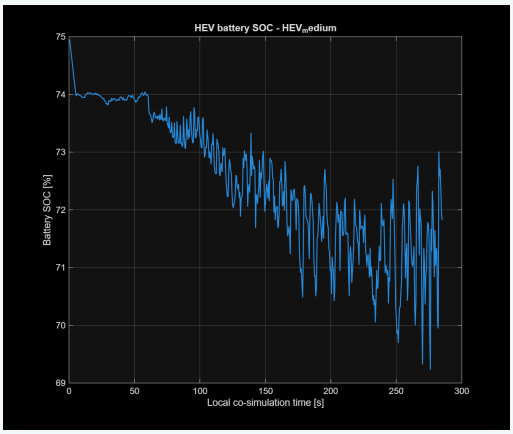
VISUAL EVIDENCE · AUTOMOTIVE SIMULATION · PYTHON

EV / HEV Traffic Co-Simulation | Rome

Original images, models and results from the supplied project material



EV/HEV simulation speed tracking generated from archived results.



HEV battery state-of-charge profile from the supplied simulation files.

RESULTS & ENGINEERING INTERPRETATION

The supplied archive contains SUMO inputs, Python scripts, Simulink model files, scenario CSV tables and plotted results. Archived outputs were inspected but not rerun.